

SOLAPUR, India

WMO: 431170

Lat: 17.628N

Lon: 75.935E

Elev: 483

StdP: 95.66

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	15.3	16.7	2.7	4.9	26.6	4.4	5.5	26.1	2.5	25.7	2.4	25.7	0.4	50	0.299

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	12.6	41.3	22.3	40.3	22.3	39.3	22.2	26.5	33.0	25.8	32.0	25.3	31.4	1.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	21.3	29.1	24.4	20.5	28.4	23.9	19.9	27.9	86.1	33.2	83.0	32.0	80.9	31.3	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 3.0	2.5	2.3	DB	12.3	43.0	1.3	0.8	11.4	43.5	10.6	44.0	9.9	44.4	8.9	44.9	(4)
(5)			WB	8.7	28.4	0.9	1.0	8.0	29.1	7.4	29.8	6.9	30.3	6.2	31.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	24.5	27.2	30.5	33.2	33.3	29.6	27.6	27.2	27.3	27.2	25.7	24.2	(6)
(7)		DBStd	3.32	1.64	1.75	2.06	1.78	1.83	2.13	1.51	1.28	1.32	1.43	1.47	1.68	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6616	448	481	636	695	722	588	546	534	519	533	472	441	(10)
(11)		CDD18.3	3574	189	248	378	445	464	338	288	276	269	275	222	183	(11)
(12)		CDH23.3	42683	1737	2791	5323	6946	7145	4152	2893	2604	2543	2754	2141	1654	(12)
(13)	CDH26.7	22484	707	1439	3220	4622	4721	2084	1080	923	956	1194	901	638	(13)	
(14)	Wind	WSAvg	0.9	0.7	0.8	0.8	0.9	1.1	1.2	1.2	1.0	0.9	0.7	0.6	(14)	
(15)	Precipitation	PrecAvg	716	2	8	5	11	31	107	128	114	199	88	26	5	(15)
(16)		PrecMax	1106	28	51	31	48	129	257	277	244	463	277	167	35	(16)
(17)		PrecMin	306	0	0	0	0	1	18	24	0	20	0	0	0	(17)
(18)		PrecStd	209	6	15	8	14	34	54	68	75	122	81	39	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.9	37.1	40.5	42.3	42.7	40.2	35.0	33.9	34.6	35.3	33.9	33.4	(19)
(20)			MCWB	19.9	20.1	21.4	22.3	22.8	23.6	24.0	23.8	23.5	22.1	20.6	20.6	(20)
(21)		2%	DB	32.5	35.8	39.3	41.3	41.6	38.0	33.6	32.9	33.3	34.0	32.7	31.9	(21)
(22)			MCWB	19.5	19.9	21.5	22.1	22.6	24.0	23.9	23.7	23.2	21.5	20.4	19.8	(22)
(23)		5%	DB	31.5	34.7	38.3	40.4	40.5	36.2	32.6	32.0	32.3	33.0	31.9	30.9	(23)
(24)			MCWB	19.2	20.0	21.2	22.1	22.4	24.1	23.9	23.5	23.2	21.5	20.5	19.2	(24)
(25)		10%	DB	30.5	33.5	37.1	39.4	39.3	34.6	31.5	31.0	31.3	32.1	31.1	30.0	(25)
(26)			MCWB	19.0	19.9	21.1	22.2	22.5	24.1	23.8	23.6	23.3	21.5	20.3	19.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.9	25.9	27.2	27.2	27.3	27.0	26.3	26.3	25.9	25.8	24.8	23.1	(27)
(28)			MCDWB	30.1	32.4	37.4	38.0	34.9	32.7	30.5	30.1	30.2	29.2	30.4	28.7	(28)
(29)		2%	WB	21.6	23.8	24.7	25.6	26.0	26.1	25.5	25.5	25.2	25.1	23.7	22.1	(29)
(30)			MCDWB	28.8	31.6	35.1	35.7	33.6	32.2	29.9	29.5	29.7	28.9	29.2	28.4	(30)
(31)		5%	WB	20.7	22.2	23.2	24.7	25.3	25.5	24.9	24.9	24.7	24.5	22.9	21.4	(31)
(32)			MCDWB	28.6	31.2	34.4	35.3	33.3	31.8	29.2	29.0	29.3	28.6	28.4	28.1	(32)
(33)		10%	WB	20.0	21.1	22.2	23.9	24.7	25.0	24.4	24.3	24.3	24.0	22.3	20.6	(33)
(34)			MCDWB	28.4	30.6	33.4	34.7	33.2	31.4	29.0	28.5	28.8	28.3	27.6	27.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.5	12.8	13.0	12.6	12.2	9.0	7.2	7.3	7.8	9.8	11.5	12.6	(35)
(36)			MCDBR	13.5	13.9	14.0	13.3	13.2	11.5	9.1	9.0	9.4	11.7	12.6	13.4	(36)
(37)			MCWBR	5.9	5.8	5.6	4.4	3.7	3.0	2.5	2.5	2.8	3.7	4.7	5.7	(37)
(38)		5% WB	MCDWB	11.4	12.1	12.7	11.9	11.7	9.1	7.3	7.3	7.7	7.8	8.9	10.4	(38)
(39)			MCWBR	5.8	6.7	6.4	5.2	4.0	3.2	2.8	3.1	3.1	3.3	4.0	5.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.447	0.462	0.535	0.596	0.588	0.558	0.534	0.518	0.507	0.535	0.487	0.452	(40)	
(41)		taud	2.095	2.079	1.902	1.815	1.851	1.980	2.058	2.105	2.131	1.992	2.061	2.103	(41)	
(42)		Ebn at Noon	830	840	791	744	741	756	776	794	803	764	785	810	(42)	
(43)		Edh at Noon	153	163	201	220	210	182	169	162	157	175	157	148	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.02	5.85	6.46	6.80	6.82	5.20	4.27	4.33	4.78	5.10	4.89	4.79	(44)	
(45)		RadStd	0.18	0.20	0.30	0.24	0.25	0.43	0.37	0.24	0.34	0.25	0.29	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+0.54	+0.47	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page